2.1 Classwork: Right-Angled Triangle Trigonometry — Markscheme

1. [Maximum mark: 11]

(a) hypotenuse 17, opposite 15, adjacent 8 **A1A1**

A1 for the hypotenuse, A1 for both the opposite and the adjacent sides. Accept the sides named by their vertices (AB, BC, AC) with their lengths.

(b) $\sin\theta = \frac{15}{17}, \quad \cos\theta = \frac{8}{17}, \quad \tan\theta = \frac{15}{8}$ **A1A1A1**

FT from (a).

(c) $\theta = \tan^{-1}\left(\frac{15}{8}\right)$ **M1**

$\theta = 61.9°$ $\quad (61.9275\ldots)$ **A1**

Accept $\sin^{-1}\left(\frac{15}{17}\right)$ *or* $\cos^{-1}\left(\frac{8}{17}\right)$ *for the M1. A value from a calculator in radian mode (*1.08*) earns M1A0.*

(d) $\sin 35° = \frac{QR}{12}$ **M1**

$QR = 12\sin 35°$

$QR = 6.88\,\text{cm}$ $\quad (6.88291\ldots)$ **A1**

(e) $\cos 50° = \frac{7}{x}$ **M1**

$x = \frac{7}{\cos 50°}$

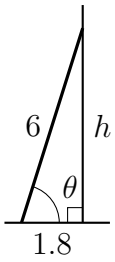
$x = 10.9$ $\quad (10.8900\ldots)$ **A1**

The common error is $x = 7\cos 50° = 4.50$*, which treats* x *as the adjacent side: M0A0.*

2. [Maximum mark: 10]

(a) $\cos\theta = \frac{1.8}{6}$ **M1**

$\theta = 72.5°$ $(72.5423\ldots)$ **A1**



(b) $h = 6\sin 72.5423\ldots°$ **M1**

$h = 5.72\,\text{m}$ $(5.72363\ldots)$ **A1**

Accept Pythagoras, $h = \sqrt{6^2 - 1.8^2}$*, for the M1. The rounded angle gives the same value,* $6\sin 72.5° = 5.72$*. FT from (a).*

(c) $\tan 28° = \frac{BT}{40}$ **M1**

$BT = 40\tan 28°$

$BT = 21.3\,\text{m}$ $(21.2683\ldots)$ **A1**

(d) $\cos 28° = \frac{40}{PT}$ **M1**

$PT = \frac{40}{\cos 28°}$

$PT = 45.3\,\text{m}$ $(45.3028\ldots)$ **A1**

Accept $\sin 28° = \frac{BT}{PT}$*, or Pythagoras* $PT = \sqrt{40^2 + BT^2}$*, for the M1. FT from (c).*

(e) angle of depression = angle TCB (alternate angles)

$\tan\alpha = \frac{21.2683\ldots}{60}$ **M1**

$\alpha = 19.5°$ $(19.5180\ldots)$ **A1**

Accept $19.5°$ *from* $\tan^{-1}\left(\frac{21.3}{60}\right) = 19.544\ldots$*. FT from (c). The common error is the angle* BTC *at the top of the tower,* $70.5°$*: M1A0.*